

of the previous block. The latter is what makes the blockchain amazingly reliable.

Merkle tree

In the Merkle tree, each non-leaf node is the hash of the values of its child nodes. By using Merkle trees, searching a transaction becomes really quick. Instead of linearly searching the transactions, the Merkle tree can decide faster if a transaction is contained in that block or not. Merkle trees are able to do this because of their structure.

Smart contracts

“Smart contracts work on the if-then promise,” to put it in Jeff Gazik’s words. As mentioned, the blockchain eliminates the need for a third party to settle transactions. While Bitcoin is like an alternative to regular money and just a distributed ledger, Ethereum has been developed as a platform that facilitates peer-to-peer contracts and applications through its own currency token. While Ethereum updates the network, it also sends some code known as the smart contract, which runs on node machines with the help of EVMs (Ethereum Virtual Machines).

Smart contracts serve the same purpose as the legal contracts in the real world. In typical cases that involve two parties, both sides have to agree on some terms to fulfil the agreement and only then sign the contract. Smart contracts do the same thing using algorithms. These contracts help to exchange services, buy things, etc, without a middleman in between. Everything in the market becomes transparent and efficient, and eliminates the chance of fraudulent activities.

In the real world, the supply chain is hampered by paper forms that have to be passed through many channels, exposing them to loss and fraud. If used, smart contracts can link up the whole supply chain by being executed at each step of the process. Also, when terms are fulfilled, the money can be transferred to the authorised party automatically, eliminating any fraudulent transactions. With the help of the blockchain, each step of the supply chain can become

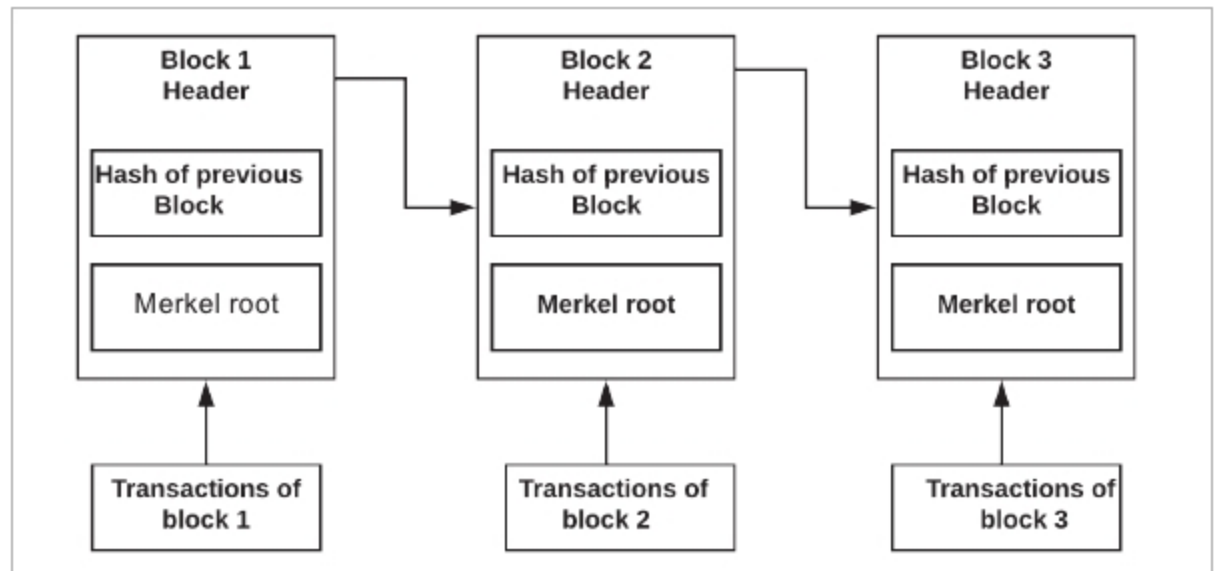


Figure 2: Structure of a block in a blockchain

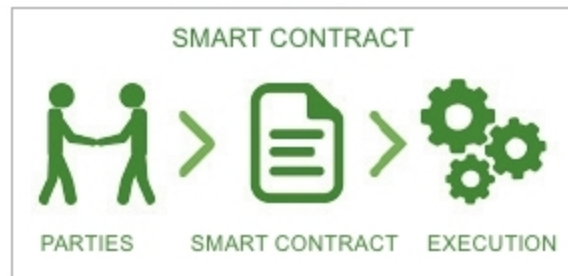


Figure 3: Smart contract

digitally accessible and transparent for all the parties on the network.

Problems with traditional databases

Traditional databases use a client-server architecture. A centralised administrator server collects data from clients and stores it at a central point. Anybody with sufficient access to the server can corrupt or destroy the data within. As a result, the moment you entrust your data to a regular database, you also become dependent on the human organisation in which the data resides. What if the company with terabytes of user data uses it for other purposes? Also, the central server can be brought down, leading to losses and a failure of the system. In a traditional database, a client can perform functions like create, read, update and delete.

Solutions using the blockchain

The core value of a blockchain is enabling a database to be directly shared across boundaries of trust, without requiring a central administrator. Blockchain consists of several decentralised nodes and all of them verify new transactions and update their

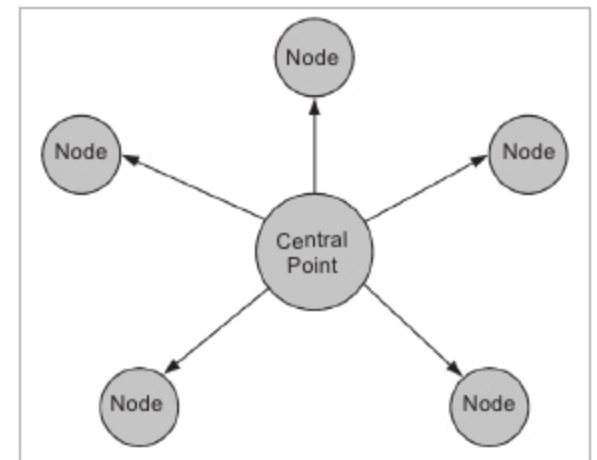


Figure 4: Centralised database — all nodes on the network are connected to a central server for accessing any data

ledgers. Pools of computing machines try solving a puzzle faster than others, after which the winner’s sequence of transactions gets added to the main chain. For solving each puzzle, the miner is rewarded with incentives in the form of tokens. After the addition of a new block on the blockchain, every node updates its ledger transaction sequence and reaches a consensus. This consensus guarantees the security of the network, making it difficult to tamper with.

Each newly added block is signed with a digital signature confirming that the majority of nodes have accepted it. This cryptographic hash digital signature maintains the integrity of data. The digital signature ensures data is unaltered and is thus reliable.

A second benefit is extreme fault tolerance because of the decentralised nature of the blockchain. The blockchain ensures that nodes that went down can always catch up on transactions they missed.